

PHYS505 Recitation — Feb 16, 2026

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Problem 3.12

Infinite grounded plane conducting sheet with thin insulating ring of radius a in the center with the central disc at potential V .

(a) Find an integral expression involving Bessel functions in cylindrical coordinates for the potential above the plane

(b) Show that the potential on the central axis ($\rho = 0$) is

$$\Phi_0 = V \left(1 - \frac{z}{\sqrt{z^2 + a^2}} \right)$$

(c) Show that the potential above the ring ($\rho = a$) is

$$\Phi_a(z) = \frac{V}{2} \left[1 - \frac{kz}{\pi a} K(k) \right], \quad k = \frac{2a}{\sqrt{z^2 + 4a^2}}$$

where $K(k)$ is the complete elliptic integral of the 1st kind.

The general solution in cylindrical coordinates looks like

$$\Phi(\rho, \phi, z) = R(\rho) Q(\phi) Z(z)$$

If we choose the constants such that the azimuthal part is $e^{\pm i\nu\phi}$ with integer ν and the axial part is $e^{\pm kz}$, the radial part will involve Bessel functions $J_\nu(k\rho)$, $N_\nu(k\rho)$. In addition,

- since there's azimuthal symmetry in the problem, $\nu = 0$
- the solution should be finite at $z \rightarrow \infty$ which rules out e^{+kz}
- the solution should also be well-behaved at $\rho \rightarrow 0$, so we can't have N_ν
- the $\nu = 0$ case also admits solutions with Z linear in z which blow up at $z \rightarrow \infty$ and R proportional to $\ln \rho$ which blow up at $\rho \rightarrow 0$
- there is no boundary condition at fixed ρ above the plane, so the range of allowable values for k is continuous and we have a Fourier integral instead of a discrete sum

(a) From the observations above,

$$\Phi(\rho, z) = \int_0^\infty A(k) J_0(k\rho) e^{-kz} dk$$

where $A(k)$ can be found using the orthogonality relation for Bessel functions

$$\int_0^\infty x J_m(kx) J_m(k'x) dx = \frac{1}{k} \delta(k - k')$$

and our boundary condition on the surface is

$$\Phi_S(\rho) = \Phi(\rho, z = 0) = \begin{cases} V, & \rho < a \\ 0, & \rho > a \end{cases}$$

so,

$$\begin{aligned} \Phi_S(\rho) &= \int_0^\infty dk' A(k') J_0(k'\rho) \\ \langle k | \Phi_S \rangle &= \int_0^\infty \Phi_S(\rho) J_0(k\rho) \rho d\rho \\ &= \int_0^\infty dk' A(k') \int_0^\infty J_0(k'\rho) J_0(k\rho) \rho d\rho \\ &= \int_0^\infty dk' A(k') \frac{\delta(k - k')}{k} = \frac{A(k)}{k} \\ A(k) &= k \int_0^\infty \Phi_S(\rho) J_0(k\rho) \rho d\rho \\ &= V k \int_0^a J_0(k\rho) \rho d\rho \end{aligned}$$

We can use the derivative formula for Bessel functions, or just the series expansion

$$\begin{aligned} J_\nu(x) &= \sum_{j=0}^\infty \frac{(-1)^j}{j! \Gamma(j + \nu + 1)} \left(\frac{x}{2}\right)^{2j+\nu} \\ \int_0^x J_0(u) u du &= \sum_{j=0}^\infty \frac{(-1)^j}{j! \Gamma(j + 1)} \int_0^x 2 \left(\frac{u}{2}\right)^{2j+1} 2d\left(\frac{u}{2}\right) \\ &= \sum_{j=0}^\infty \frac{(-1)^j}{j! \Gamma(j + 1)} \frac{4}{2j + 2} \left(\frac{x}{2}\right)^{2j+2} \\ \Gamma(j + 2) &= (j + 1) \Gamma(j + 1) \\ \int_0^x J_0(u) u du &= x \sum_{j=0}^\infty \frac{(-1)^j}{j! \Gamma(j + 2)} \left(\frac{x}{2}\right)^{2j+1} = x J_1(x) \\ k \int_0^a J_0(u) u du &= k \frac{1}{k^2} \int_0^{ka} J_0(k\rho) (k\rho) d(k\rho) = \frac{1}{k} ka J_1(ka) \\ A(k) &= V a J_1(ka) \end{aligned}$$

and the solution can be found by inserting this into the integral formula for Φ ,

$$\Phi(\rho, z) = Va \int_0^\infty J_1(ka) J_0(k\rho) e^{-kz} dk$$

Before diving into the math, let's stare at this formula and see if it makes sense:

- it is dimensionally correct since (ak) is dimensionless and so are the Bessel functions;
- for large k , the Bessel functions decay as $1/\sqrt{k}$, so the integrand decays faster than exponential and the integral is finite for non-negative values of z ;
- due to the e^{-kz} factor in the integrand, the potential decays to 0 as expected for large positive z since we have an infinite grounded plane and a compact region with non-zero potential near the origin;
- at the origin, the boundary condition is satisfied

$$\Phi(0, 0) = Va \int_0^\infty J_1(ka) dk = V \int_0^\infty J_1(u) du = V$$

since the Bessel functions are normalized to 1 (not surprising since we used the potential on the boundary to derive the expression but it's an additional check);

- in the limit $a \rightarrow 0$, the disc disappears and the potential above the full infinite grounded plane should be 0 everywhere which follows from this formula since $J_1(0) = 0$.

(b) For the potential on the z -axis,

$$\begin{aligned} \Phi_0(z) \equiv \Phi(\rho = 0, z) &= Va \int_0^\infty J_1(ka) J_0(0) e^{-kz} dk \\ &= V \int_0^\infty J_1(u) e^{-uz/a} du \end{aligned}$$

The way I evaluate this integral would probably make mathematicians cringe but as Boltzmann supposedly said "Eleganz sei die Sache der Schuster und Schneider" (elegance should be left to shoemakers and tailors), so here it is – starting with

$$\begin{aligned} I(\alpha) &= \int_0^\infty e^{-\alpha u} du = \frac{1}{\alpha} \\ \left(\frac{d}{d\alpha} \right)^n I(\alpha) &= (-1)^n \int_0^\infty u^n e^{-\alpha u} du = (-1)^n n! \alpha^{-(n+1)} \\ \int_0^\infty u^n e^{-\alpha u} du &= \frac{n!}{\alpha^{n+1}}, \quad n \geq 0 \\ \int_0^\infty u^n e^{-u/\alpha} du &= n! \alpha^{n+1} \end{aligned}$$

define

$$\begin{aligned}
S(\alpha) &\equiv \int_0^\infty J_1(u) e^{-u/\alpha} du \\
&= \sum_{j=0}^\infty \frac{(-1)^j}{j!(j+1)!} \int_0^\infty \left(\frac{u}{2}\right)^{2j+1} e^{-u/\alpha} du \\
&= \sum_{j=0}^\infty \frac{(-1)^j}{j!(j+1)!} \frac{1}{2^{2j+1}} (2j+1)! \alpha^{2j+2} \\
&= - \sum_{j=1}^\infty \frac{(-1)^j}{(j-1)!} \frac{(2j-1)!}{2^{2j-1}} \frac{(\alpha^2)^j}{j!}
\end{aligned}$$

which looks messy but examining the terms carefully

$$\begin{aligned}
\frac{(2j-1)!}{2^{2j-1}} &= \frac{1}{2} \frac{2}{2} \frac{3}{2} \frac{4}{2} \cdots \frac{2j-2}{2} \frac{2j-1}{2} = \left(\frac{1}{2} \frac{3}{2} \cdots \frac{2j-1}{2} \right) (j-1)! \\
\frac{(-1)^j}{(j-1)!} \frac{(2j-1)!}{2^{2j-1}} &= \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) \left(-\frac{5}{2} \right) \cdots \left(-\frac{2j-1}{2} \right) \\
&= \left(-\frac{1}{2} \right) \left(-\frac{1}{2} - 1 \right) \left(-\frac{1}{2} - 2 \right) \cdots \left[-\frac{1}{2} - (j-1) \right]
\end{aligned}$$

we see that it looks like the binomial series

$$(1 + \alpha^2)^{-1/2} = \sum_{j=0}^\infty \left(-\frac{1}{2} \right) \left(-\frac{1}{2} - 1 \right) \left(-\frac{1}{2} - 2 \right) \cdots \left[-\frac{1}{2} - (j-1) \right] \frac{(\alpha^2)^j}{j!}$$

except for the extra term with $j = 0$ and the overall minus sign

$$S(\alpha) = - \left[(1 + \alpha^2)^{-1/2} - 1 \right]$$

Therefore,

$$\begin{aligned}
\int_0^\infty J_1(u) e^{-u/\alpha} du &= 1 - (1 + \alpha^2)^{-1/2} \\
\Phi_0(z) &= V \left\{ 1 - \left[1 + \left(\frac{a}{z} \right)^2 \right]^{-1/2} \right\} \\
\Phi(\rho = 0, z) &= V \left(1 - \frac{z}{\sqrt{z^2 + a^2}} \right), \quad z > a
\end{aligned}$$

Note that this only holds for $z > a$, where the binomial series converges. For $z < a$, you can make the substitution $\alpha \rightarrow 1/\alpha$ to find the same result. Another line of attack would be to use the identity in problem 3.16c (which can also be

derived by term-by-term integration of the series as above)

$$\begin{aligned}
I(\rho, z) &\equiv \int_0^\infty e^{-kz} J_0(k\rho) dk = \frac{1}{\sqrt{\rho^2 + z^2}}, \quad z \geq 0 \\
\partial_z I(\rho, z) &= - \int_0^\infty k e^{-kz} J_0(k\rho) dk = - \frac{z}{(\rho^2 + z^2)^{3/2}} \\
\int_0^\infty k e^{-kz} J_0(ka) dk &= \frac{z}{(a^2 + z^2)^{3/2}}
\end{aligned}$$

and the other identity we derived above from the series formula for the Bessel function

$$\begin{aligned}
\frac{d}{dx} J_0(x) &= -J_1(x) \\
d[J_0(ka)] &= -a J_1(ka) dk
\end{aligned}$$

Integrating by parts,

$$\begin{aligned}
\int u dv &= uv - \int v du \\
\frac{\Phi_0(z)}{Va} &= \int_0^\infty e^{-kz} J_1(ka) dk = \int_0^\infty e^{-kz} d\left[-\frac{J_0(ka)}{a}\right] \\
&= \left[-e^{-kz} \frac{J_0(ka)}{a}\right]_0^\infty + \int_0^\infty \frac{J_0(ka)}{a} d[e^{-kz}] \\
&= \frac{1}{a} - \frac{z}{a} \int_0^\infty e^{-kz} J_0(ka) dk
\end{aligned}$$

where we have used the fact that $J_0(ka)$ decays as $ka \rightarrow \infty$. Using $I(a, z)$ for the remaining integral,

$$\begin{aligned}
\frac{\Phi_0(z)}{Va} &= \frac{1}{a} - \frac{z}{a} \frac{1}{\sqrt{\rho^2 + z^2}} \\
\Phi_0(z) &= V \left(1 - \frac{z}{\sqrt{z^2 + a^2}}\right)
\end{aligned}$$

The result has the correct values at the boundaries: V at $z = 0$ and 0 as $z \rightarrow \infty$.

(c) For $\rho = a$

$$\begin{aligned}
\Phi_a(z) &\equiv \Phi(\rho = a, z) = Va \int_0^\infty J_1(ka) J_0(ka) e^{-kz} dk \\
&= V \int_0^\infty J_1(u) J_0(u) e^{-uz/a} du
\end{aligned}$$

which we are supposed to transform to an elliptic integral. While this can be done with “straightforward” algebra/calculus (using the complex integral representation of the Bessel function in problem 3.16d, for example), the end result

isn't particularly illuminating, so I won't bore you with the details. Instead, having seen a few useful identities, let's check the behavior of the electric field.

Starting from the general expression in part (a)

$$\begin{aligned}
\Phi(\rho, z) &= V a \int_0^\infty J_1(ka) J_0(k\rho) e^{-kz} dk \\
E_z &= -\partial_z \Phi = V a \int_0^\infty J_1(ka) J_0(k\rho) e^{-kz} k dk \\
E_\rho &= -\partial_\rho \Phi = -V a \int_0^\infty J_1(ka) \left[\frac{d}{d\rho} J_0(k\rho) \right] e^{-kz} dk \\
&= -V a \int_0^\infty J_1(ka) \left[-k \frac{d}{d\rho} J_1(k\rho) \right] e^{-kz} dk \\
&= V a \int_0^\infty J_1(ka) J_1(k\rho) e^{-kz} k dk
\end{aligned}$$

we observe the following:

- On the conducting plane, we expect the radial component of the field to vanish everywhere except at the ring where the potential has a jump discontinuity

$$\begin{aligned}
E_\rho(\rho, z=0) &= V a \int_0^\infty J_1(ka) J_1(k\rho) k dk \\
&= V \delta(\rho - a) \\
\Phi(\rho = a^-, 0) - \Phi(\rho = a^+, 0) &= \int_{a^-}^{a^+} E_\rho d\rho = V
\end{aligned}$$

- while the axial component should be pointing up inside the ring and down outside

$$E_z(\rho, z=0) = V a \int_0^\infty J_1(ka) J_0(k\rho) k dk$$

which doesn't seem obvious from this expression in which the integrand oscillates wildly.

- Above the ring, we expect the radial component of the electric field to point outward

$$E_\rho(\rho = a, z) = V a \int_0^\infty [J_1(ka)]^2 e^{-kz} k dk$$

which is clear from this expression since each of the three factors in the integrand is +ve.